

Lab Cycle 1

Aim: Calculator Application:

- a) Using with and without R objects on console.

Program:

With R Objects:

```
print("Welcome to Calculator (With R Objects)")
```

```
operator <- readline("Enter operator (+, -, *, /): ")
num1 <- as.numeric(readline("Enter first number: "))
num2 <- as.numeric(readline("Enter second number: "))
if (operator == "+") {
  result <- num1 + num2
  print(paste("Result =", result))
} else if (operator == "-") {
  result <- num1 - num2
  print(paste("Result =", result))
} else if (operator == "*") {
  result <- num1 * num2
  print(paste("Result =", result))
} else if (operator == "/") {
  if (num2 != 0) {
    result <- num1 / num2
    print(paste("Result =", result))
  } else {
    print("Error: Division by zero")
  }
} else {
  print("Invalid operator")
}
```

Output:

```
Welcome to Calculator
Enter operator(+,-,*,/): +
Enter first number:3
Enter second number:8
Result=11
```

Without R Objects:

```
cat("Simple Calculator 501(without object)\n")
cat("Enter first number: ")
num1 <- as.numeric(readline())

cat("Enter operator (+, -, *, /): ")
```

```
op <- readline()

cat("Enter second number: ")
num2 <- as.numeric(readline())

if (op == "+") {
  cat("Result:", num1 + num2, "\n")
} else if (op == "-") {
  cat("Result:", num1 - num2, "\n")
} else if (op == "*") {
  cat("Result:", num1 * num2, "\n")
} else if (op == "/") {
  if (num2 == 0) {
    cat("Error: Division by zero!\n")
  } else {
    cat("Result:", num1 / num2, "\n")
  }
} else {
  cat("Invalid operator\n")
}
```

Output:

Simple Calculator

Enter operator(+,-,*,/): +

Enter first number:3

Enter second number:8

Result=11

Aim: Calculator Application:

b) Using mathematical functions on console.

Program:

```
num1<-as.numeric(readline("Enter first number: "))
num2<-as.numeric(readline("Enter Second number: "))
abs(num1)
sqrt(num2)
round(num1,3)
ceiling(num1)
floor(num1)
sin(num2)
cos(num2)
tan(num2)
log(num2)
exp(num2)
factorial(num2)
choose(5,2)
sign(num1)
```

Output:

```
> num1<-as.numeric(readline("Enter first number: "))
Enter first number:
> num2<-as.numeric(readline("Enter Second number: "))
Enter Second number:
>
> abs(num1)
[1] NA
> sqrt(num2)
[1] NA
> round(num1,3)
[1] NA
> ceiling(num1)
[1] NA
> floor(num1)
[1] NA
> sin(num2)
[1] NA
> cos(num2)
[1] NA
> tan(num2)
[1] NA
> log(num2)
```

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```
[1] NA  
> exp(num2)  
[1] NA  
> factorial(num2)  
[1] NA  
> choose(5,2)  
[1] 10  
> sign(num1)  
[1] NA  
>
```

Aim: : Calculator Application:

c) Write an R script, to create R objects for calculator application and save in a specified location in disk.

Program:

```
operator <- readline("Enter operator (+, -, *, /): ")
num1 <- as.numeric(readline("Enter first number: "))
num2 <- as.numeric(readline("Enter second number: "))
if(op==""){
  res<-num1+num2
}else if(op=="-"){
  res<-num1-num2
}else if(op=="*"){
  res<-num1*num2
}else if(op=="/"){
  res<-num1/num2
}else{
  res<-NA
  cat("Invalid Operator")
}
Calculator_data<-list(
  number1=num1,
  number2=num2,
  operation=op,
  result=res
)
save_path<-"C:\Users\admin\Desktop\503\503\data.RData"
save(calculator_data,file=sava_path)
cat("data save successfully to",save_path,"\n")
```

Output:

```
Enter num1:15
Enter num2:20
Enter operator(+,-,*,/): +
Result:35
```

Aim: : Descriptive Statistics:

a) Write an R script to find basic descriptive statistics using summary.

Program:

```
myData <- read.csv("fitness_exercises.csv", stringsAsFactors = FALSE)
print(head(myData))
print(summary(myData))
```

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Output:

```
> # Import the data using read.csv()
> myData <- read.csv("fitness_exercises.csv", stringsAsFactors = FALSE)

> # Print the first 6 rows
> print(head(myData))
  bodyPart equipment gifUrl id
1 waist body weight http://d205bpvrqc9yn1.cloudfront.net/0001.gif 1
2 waist body weight http://d205bpvrqc9yn1.cloudfront.net/0002.gif 2
3 waist body weight http://d205bpvrqc9yn1.cloudfront.net/0003.gif 3
4 upper legs body weight http://d205bpvrqc9yn1.cloudfront.net/1512.gif 1512
5 waist body weight http://d205bpvrqc9yn1.cloudfront.net/0006.gif 6
6 back cable http://d205bpvrqc9yn1.cloudfront.net/0007.gif 7

  name target
1 3/4 sit-up abs
2 45° side bend abs
3 air bike abs
4 all fours squad stretch quads
5 alternate heel touchers abs
6 alternate lateral pulldown lats

> # Use summary() to get descriptive statistics
> print(summary(myData))
  bodyPart equipment gifUrl id
Length:1327 Length:1327 Length:1327 Min. : 1.0
Class :character Class :character Class :character 1st Qu.: 378.5
Mode :character Mode :character Mode :character Median : 862.0
                                     Mean :1189.5
                                     3rd Qu.:1661.5
                                     Max. :5201.0

  name target
Length:1327 Length:1327
Class :character Class :character
Mode :character Mode :character
```

Aim: : Descriptive Statistics:

b) Write an R script to find subset of dataset by using subset

Program:

```
df <- data.frame(  
  age = 1:10,  
  category=rep(c("A","B"),each=5),  
  value=rnorm(10)  
)
```

```
# Print the original data frame  
print("Original Data Frame:")  
print(df)
```

```
# 1. Subset based on a condition (e.g., age > 27)  
subset_age <- subset(df, age>2)  
print("Subset (age > 2):")  
print(subset_age)
```

```
# 2. Subset based on multiple conditions (e.g., age > 27 and city is 'London')  
subset_age_value <- subset(df, age > 2 & value>1)  
print("Subset (age > 2 & value > 1):")  
print(subset_age_value)
```

Output:

```
# Print the original data frame  
> print("Original Data Frame:")  
[1] "Original Data Frame:"
```

```
> print(df)  
  age category  value  
1  1      A 0.75824488  
2  2      A -0.89647431  
3  3      A -0.60503366  
4  4      A 0.18010395  
5  5      A 0.01542015  
6  6      B -0.29360162  
7  7      B 0.28459992  
8  8      B -1.02765601  
9  9      B -0.18700653  
10 10     B 0.67524209
```

```
> # 1. Subset based on a condition (e.g., age > 27)
```

```
> subset_age <- subset(df, age>2)
```

```
> print("Subset (age > 2):")
```

```
[1] "Subset (age > 2):"
```

```
> print(subset_age)
```

	age	category	value
3	3	A	-0.60503366
4	4	A	0.18010395
5	5	A	0.01542015
6	6	B	-0.29360162
7	7	B	0.28459992
8	8	B	-1.02765601
9	9	B	-0.18700653
10	10	B	0.67524209

```
> # 2. Subset based on multiple conditions (e.g., age > 27 and city is 'London')
```

```
> subset_age_value <- subset(df, age > 2 & value>1)
```

```
> print("Subset (age > 2 & value > 1):")
```

```
[1] "Subset (age > 2 & value > 1):"
```

```
> print(subset_age_value)
```

```
[1] age    category value
```

```
<0 rows> (or 0-length row.names)
```

Aim: Reading and Writing Different Types of Datasets:

a) Reading different types of data sets (.txt, .csv) from web and disk and writing in file in specific disk location

Program:

```
csv_data <- read.csv("C:/Users/admin/Desktop/503/503/industry.csv")
head(csv_data)
new_row <- data.frame(Industry="Healthcare")
csv_data<-rbind(csv_data,new_row)
head(csv_data)
txt_data <- readLines("C:/Users/admin/Desktop/503/503/text.txt")
cat(txt_data)
writeLines("Computer Engineering", file="C:/Users/admin/Desktop/503/503/text.txt",append=TRUE)
cat(txt_data)
```

Output:

```
> csv_data <- read.csv("C:/Users/admin/Desktop/503/503/industry.csv")
> head(csv_data)
  Industry
1  Technology
2   Finance
3    Retail
4 Manufacturing
> new_row <- data.frame(Industry="Healthcare")
> csv_data<-rbind(csv_data,new_row)
> head(csv_data)
  Industry
1  Technology
2   Finance
3    Retail
4 Manufacturing
5  Healthcare
> txt_data <- readLines("C:/Users/admin/Desktop/503/503/text.txt")
> cat(txt_data)
Electrical Engineering
Mechanical Engineering
> writeLines("Computer Engineering", file="C:/Users/admin/Desktop/503/503/text.txt",append=TRUE)
> cat(txt_data)
```

Electrical Engineering

Mechanical Engineering

Aim: Reading and Writing Different Types of Datasets:

b)Reading Excel data sheet in R.

Program:

```
library(readxl)
file_path <- "C:/Users/admin/Desktop/503/503/excel.xls"
data <- read_excel(file_path)
print(head(data))
```

Output:

```
> print(results)
  name age      email      title      author
1 John Doe 30 john.doe@example.com <NA> <NA>
2 Jane Smith 25 jane.smith@example.com <NA> <NA>
3 <NA> <NA> <NA> The Adventure Begins Robert Johnson
  year
1 <NA>
2 <NA>
3 2022
```

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Aim: Reading and Writing Different Types of Datasets:

c)Reading XML dataset in R

Program:

```
library("XML")  
library("methods")  
results <- xmlToDataFrame("file:///C:/Users/admin/Desktop/503/503/xmlFile.xml")  
print(results)
```

Output:

	`0`	`First Name`	`Last Name`	Gender	Country	Age	Date	Id	
		<i><dbl></i>	<i><chr></i>	<i><chr></i>	<i><chr></i>	<i><chr></i>	<i><dbl></i>	<i><chr></i>	<i><dbl></i>
1	1	Dulce	Abril	Female	United States	32	15/10/2017	1562	
2	2	Mara	Hashimoto	Female	Great Britain	25	16/08/2016	1582	
3	3	Philip	Gent	Male	France	36	21/05/2015	2587	
4	4	Kathleen	Hanner	Female	United States	25	15/10/2017	3549	
5	5	Nereida	Magwood	Female	United States	58	16/08/2016	2468	
6	6	Gaston	Brumm	Male	United States	24	21/05/2015	2554	